

COPELAND ROYALL

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copey.dev

SKILLS

Languages Python, TypeScript, Go

Software AWS, GCP, Terraform, Terragrunt, Pulumi, Kubernetes, GitHub Actions, Flux CD, Ansible, Docker

EXPERIENCE

Freelance Software Engineer

January 2025 – present

Opinion Health

Architected and built a multi-region pharmaceutical research AWS environment using Terraform and Terragrunt

- Designed account-isolated RDS Aurora architecture to enforce per-region data residency
- Built org-wide AWS security governance model with dedicated log-archive account for CloudTrail and pgAudit (Lambda export, 6-year retention), dedicated security account for GuardDuty and CloudTrail administration, and SCP guardrails for region locks, encryption, and KMS deletion protection
- Implemented GitOps CI/CD pipeline in GitHub Actions with Terraform module unit tests and security/secret scanning, gating every deploy on PR review and automated drift detection
- Built Python NLP pipeline for extreme multi-label classification (1.4k labels) across 130k patient records spanning 10 years, combining traditional techniques with Bedrock embeddings and batch LLM inference; anonymized patients until database ingest to minimize exposure risk
- Restricted database access to SSO/MFA-enforced sessions from Ansible-provisioned VMs
- Eliminated Aurora cold-start query timeouts by triggering a warmup Lambda on user VM login

Outspoken Education

- Built a full-stack news intelligence system with continuous monitoring of news sources for editorial reporting
- Implemented Python NLP clustering pipeline to categorize and combine articles into stories with a 39% compression rate, finding the most significant news and reducing high-volume feed to targeted topics
- Deployed multi-processes and Postgres – with race-protected migrations – on self-managed Kubernetes cluster using GitHub Actions CI/CD pipeline and Flux CD GitOps; kube-prometheus-stack monitoring

Keygraph

- Developed multi-account, -region and -environment AWS platform module in TypeScript/Node.js Pulumi IaC
- Deployed 6 containerized production Go apps using App Runner, RDS Aurora and SOPS secrets

Historical Artist Website

- Built [website](#) for acclaimed artist with Hugo, Cloudflare Worker and R2; JavaScript OAuth proxy for editing

Visa Software Engineer

September 2023 – May 2024

Accelerated regulated payment product rollouts on GCP with TypeScript IaC and led deployment of cost monitoring in org.

- Implemented TypeScript Pulumi IaC package deploying 256 resources across 4 GCP projects, meeting PCI compliance requirements
- Wrote reusable Pulumi packages for GKE, IAM, and KMS resource management with comprehensive TypeScript unit test coverage
- Led team initiative to deploy Cloudability FinOps in 45 test and production projects; calculated 10% cost savings
- Installed and updated Kubernetes cluster addons using Argo CD, Kustomize and Helm
- Migrated Concourse CI pipelines to new load-test platform

Visa DevOps Engineering Intern

Summer 2022

Developed foundational Terraform modules for AWS RDS Aurora and ElastiCache Redis including best-practice defaults to enforce consistency across teams.

Tangible AI Natural Language Processing Intern

Summer 2021

EDUCATION

University of Edinburgh BSc Computer Science, First-class honors

2019 – 2023

Dissertation: Microservices and message queue-based multi-receiver ADS-B aircraft tracking data aggregation system written in Go, utilizing Apache Kafka, MongoDB and GraphQL, achieving 10ms end-to-end latency

PROJECTS

- Built first [3D aircraft flight heatmap and radar viewer](#) to see paths curving with the globe (instead of distorted 2D)
- Empowered Siri with local aircraft tracking, achieving over 18,500 downloads from GHCR